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Top Skills

System on a Chip (SoC)
Interconnect Architecture
AMBA Protocols

Honors-Awards

Arm Quality Award 2020 for the Mbed OS FPGA CI Test Shield
UT ECE Senior Design First Place
Engineering AI Showcase Most Innovative Award

Michael Coulter

Senior Software Engineer at Arm
Austin, Texas, United States

Summary

I am a Software Engineer working in the Systems IP group at Arm. I currently work on tooling for coherent and non-coherent interconnect IP.

Experience

Arm

7 years 10 months

Senior Software Engineer

April 2023 - Present (3 years 7 months)
Austin, Texas, United States

Working as a software engineer for the Arm Interconnect Architect, a desktop application software tool dedicated to the coherent and non-coherent interconnect product lines. The application is built with Angular/TypeScript and employs object-oriented programming for a scalable codebase. The application serves both internal engineers and external customers, enabling a structured process for building interconnect microarchitecture configurations. Notable features include a canvas tool for viewing and editing the entire interconnect system, a build tab for generating RTL from configurations, and a console for real-time user feedback. The tool utilizes a custom Python backend, with a large set of interconnect configuration APIs.

Software Engineer

April 2020 - April 2023 (3 years 1 month)
Austin, Texas, United States

- Arm graduate rotation program:
- Wrote RTL for several safety mechanisms for one of the Arm non-coherent interconnect product lines
- Worked on the Virtual IP (VIP) team running simulations and working with various VIP software blocks
- Started working on the Arm Interconnect Assistant desktop tool
- Worked on a web-application dashboard for the Arm Pelion Products group

Graduate Software Engineer

January 2019 - April 2020 (1 year 4 months)
Austin, Texas Area

- Worked on the Arm mbed FPGA test shield. The FPGA test shield was created to test all of the various IO functionality for the myriad of mbed supported development boards. I helped to create an entire verilog code base for the FPGA, as well as the FPGA test shield PCB and form factor boards in order to achieve this.
- Debugged issues and implemented fixes for Mbed OS.

Arm
Internet of Things Engineering Intern
May 2018 - August 2018 (4 months)
Austin, Texas Metropolitan Area

- Outlined and began initial work on the Mbed FPGA CI test shield project
- Worked with and made improvements to Mbed OS
- Integrated the ATECC608A cryptography chip library with Mbed I2C API to enable future hardware
encryption goal
- Developed content for Mbed OS handbook, code examples and tutorials
- Added I2C tests to CI test shield library

Arm
Systems and Software IP Engineering Intern
May 2017 - August 2017 (4 months)
Austin, Texas Metropolitan Area

- Tested and debugged an Arm Cortex-A CPU embedded logic analyzer using a UVM test bench
- Created a script to decompress trace data from the embedded logic analyzer (ELA) that was shipped to customers

Arm
ATG Verification Enablement Intern
May 2016 - August 2016 (4 months)
Austin, Texas Metropolitan Area

- Helped to implement architectural system coverage
- Wrote Python scripts to manage TPIs (architectural test suites) and test data from assembly test
generation tool
- SQLite3 database for test data management

Arm

IT Intern

October 2014 - August 2015 (11 months)

Austin, Texas Metropolitan Area

- Imaged and distributed new machines, made local admins
- Worked on service tickets including network patching and machine, phone, monitor, and desk setup
- Developed customer service, delegation, and organizational skills

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Education

The University of Texas at Austin

Electrical and Computer Engineering, Computer Architecture and Embedded Systems, Integrated Circuits · (2014 - 2018)